Vidyavardhini's College of Engineering and Technology, Vasai
Department of Computer Science & Engineering (Data Science)

Academic Year 2025-26

Assignment No. 1

Class / Sem: TE/VI **Course Code:** CSL605

Course Name: Cloud Computing

Date of Announcement: 14/01/26 **Submission Due Date:** 27/01/26

LO1: Implement different types of virtualization techniques.

Q.No.	Question	Bloom Level
Q1.	A cloud service provider wants to run multiple customer operating systems on the same physical server with strong isolation and flexible resource allocation. Apply a suitable **virtualization technique** for this scenario and explain how it would be implemented. Justify your choice.	3
Q2	A college wants to provide the same desktop environment to all students in computer labs, with centralized management and easy updates. Apply **desktop virtualization** to this scenario and describe how it can be implemented.	3

Ans 1: Suitable Technique: Server Virtualization using Hypervisor (Full Virtualization)

A cloud service provider needs to run multiple customer operating systems on the same physical server while maintaining strong isolation and flexible resource allocation. The most suitable virtualization technique for this scenario is Server Virtualization using a Hypervisor (Full Virtualization).

Explanation:

Server virtualization allows a single physical server to host multiple Virtual Machines (VMs). Each VM runs its own operating system and applications independently. A hypervisor is the software layer that sits between the hardware and the virtual machines and manages resource allocation.

There are two types of hypervisors:

1. Type 1 Hypervisor (Bare Metal) – Installed directly on hardware
   Examples: VMware ESXi, Microsoft Hyper-V, Xen
2. Type 2 Hypervisor – Installed on top of an operating system
   Examples: VirtualBox, VMware Workstation

Cloud providers typically use Type 1 hypervisors because they offer better performance and stronger isolation.

Implementation Steps

1. Install a Hypervisor
   The cloud provider installs a Type 1 hypervisor on the physical server.
2. Create Virtual Machines
   Multiple VMs are created using the hypervisor.
3. Allocate Resources
   Each VM is assigned resources such as CPU, RAM, Storage, Network bandwidth,etc.
4. Install Operating Systems
   Customers install their preferred operating systems (Linux, Windows, etc.) inside their VMs.
5. Isolation Between VMs
   The hypervisor ensures that each VM is isolated so that one customer cannot access another customer's data.
6. Dynamic Resource Allocation
   Resources can be adjusted dynamically depending on workload.

Justification

Server virtualization is suitable because:

- It provides strong isolation between customers.
- It supports multiple operating systems simultaneously.
- Resources can be scaled and managed efficiently.
- It improves hardware utilization and reduces costs.
  Therefore, server virtualization using a hypervisor is the best solution for running multiple customer operating systems securely on the same physical server.

Ans 2: Suitable Technique: Desktop Virtualization using Virtual Desktop Infrastructure (VDI)

In this scenario, a college wants to provide the same desktop environment to all students across computer labs with centralized management and easy updates. The most suitable solution is Desktop Virtualization using Virtual Desktop Infrastructure (VDI).

Explanation

Desktop virtualization allows desktop environments to run on centralized servers rather than individual computers. Students access their desktops remotely through thin clients or lab computers. In VDI, the desktop operating systems are hosted on a central server, and users connect to them over the network.

Implementation Steps

1. Set Up a Central Server
   A powerful server is installed in the college data center.
2. Install Virtualization Software
   Desktop virtualization platforms such as:
   - VMware Horizon
   - Citrix Virtual Apps and Desktops
   - Microsoft Azure Virtual Desktop
3. Create Virtual Desktops
   Multiple virtual desktops with the same configuration (OS, applications, settings) are created.
4. Centralized Management
   Administrators manage all desktops from a central control panel.
5. Student Access
   Students log in from lab computers or thin clients to access their virtual desktops.
6. Updates and Maintenance
   Software updates are performed on the central image, and all virtual desktops receive the update automatically.

Benefits

Desktop virtualization provides several advantages:

- Centralized management of all desktops
- Easy software updates and maintenance
- Consistent desktop environment for all students
- Reduced hardware requirements in labs
- Improved security and data protection

Conclusion

Using Virtual Desktop Infrastructure (VDI) allows the college to efficiently manage lab computers, maintain a consistent desktop environment, and simplify updates and administration.